Object Oriented Programming (OOP)

Lecture 1 : Course Overview

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Course Overview

- Course Topics
  - OOP Concepts
    - Data Abstraction/Encapsulation
    - Inheritance
    - Polymorphism
    - Generic Programming and STL (Standard Template Library)
  - Advanced C++
  - UML (Unified Modeling Language)
    - Modeling Object-Oriented Software Systems
- Course Objectives
  - Understand OOP concepts and apply it to real programming
  - Improve the ability of using advanced OOP features of C++
  - Conduct Team Projects (Large Software Development)
    - Design/Implementation/Documentation

Course Format

- This course consists of
  - Lectures (in a classroom on campus)
  - Programming Projects
    - 2 individual projects + 2 team-based projects
  - Presentation (in the case of team projects)

Reference Books

- No Required Textbook.
- Reference Books
  - *"An Intro. To Object Oriented Programming" , T. A. Budd*
  - *"Thinking in C++, Volume One/Two: Introduction to Standard C++(2nd Edition)", Bruce Eckel, Prentice Hall, 2000/2004*
  - Full text of "Thinking in C++" is available as pdf files
    - http://www.cs.ust.hk/~dekai/library/ECKEL_Bruce
  - *"C++ : Effective Object-Oriented Software Construction : Concepts, Practices, Industrial Strategies and Practices"*, Kayshav Dattari, 2nd Edition, 1999 (Page 1 ~ Page 574)
- Some of online materials will be used.

Evaluation

- Midterm Exam: 35%
- Final Exam: 35%
- Projects/Assignments: 25%
  - 4 programming projects
- Attendance : 5%
- You may get F grade if you miss ¼ or more of the whole classes
- This evaluation policy may change depending on COVID-19 situation.

Instructor Information

- Name : Bong-Soo Sohn
- Office : 310-738
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- Tel : 820-5843
- Please feel free to contact me for any of class issues.

Announcement

- Class Website
  - https://sohnbong.github.io/oop26
  - Class information such as lecture notes is accessible through this website
  - We will also use e-Class for the submission of homework and project results.
- Individual Programming Assignments
  - I encourage you to study and discuss together for doing programming assignments.
  - However, you must do programming YOURSELF.
  - You must not share any of source code with other students.
  - Academic dishonesty (e.g. cheating, plagiarism, and etc.) is seriously prohibited. Heavy penalty may be imposed for that.

Schedule

1	Course Overview, OOP Introduction
2	OOP Introduction, C++ Basics
3	C++ Basics
4	Object Oriented Design
5	Data Encapsulation
6	Data Encapsulation, Inheritance
7	Polymorphism
8	Midterm exam

Schedule

9	Polymorphism, Template
10	Template
11	STL (Standard Template Library)
12	STL
13	UML, Modern C++ features
14	Modern C++ features
15	Term Project Presentation
16	Final Exam

Remarks

- We will mainly use C++ in our lecture.
- Project/Programming Based Course
  - Prerequisite: C/JAVA programming, Data Structures
    - This course assumes that students have basic knowledge of C and JAVA grammars and programming ability. Basic knowledge of data structures such as array and linked list are also required.
  - We will have significant amount of programming assignments including team projects and presentation. If your programming ability is not good enough, it may be difficult to follow the class.
- English Lecture + English Student Presentation
- You should use English for reports and exams.

Projects Plan

- Make up a team
  - Deadline : 11:59pm on Sep. 10
  - Only the leader of each team needs to post an article (name/email/phone#/student-id# of team members, mark the team leader) in eClass (homework 0)
  - Expected Team Size : 4, 5, or 6

    (If you want, team size 1, 2, 3 or 7 will be also allowed.

    If your team size is large, I expect the quality of the team project result will be also high.)
- There will be 4 programming projects during the semester.

  (2 projects will be done individually, the other 2 projects will be done in team.)
- Each team will present one of project 1 , 2 , and 3.
- All teams should conduct the final term (4th) project and give presentation.

If you have difficulty in making a team..

- team with a single member (team size = 1) is also possible, if you want.
  - In this case, you also need to post an article in eclass (homework 0), stating that you are the only member in your team
    - If you don't register your team in hw 0, I will assume you belong to "size 1 team".
  - In this case, you also need to do project 1, 2, 3, 4 as follows.
    - project 1, 2, and 3 as individual project.
    - project 4 as team project
      - you need to submit source code, executable code, project report, and video file for your proj presentation (there will be no class presentation in this case)